

Emotion Regulation and Decision Making Under Risk and Uncertainty

Renata M. Heilman and Liviu G. Crişan
Babeş-Bolyai University

Daniel Houser
George Mason University

Mircea Miclea and Andrei C. Miu
Babeş-Bolyai University

It is well established that emotion plays a key role in human social and economic decision making. The recent literature on emotion regulation (ER), however, highlights that humans typically make efforts to control emotion experiences. This leaves open the possibility that decision effects previously attributed to acute emotion may be a consequence of acute ER strategies such as cognitive reappraisal and expressive suppression. In Study 1, we manipulated ER of laboratory-induced fear and disgust, and found that the cognitive reappraisal of these negative emotions promotes risky decisions (reduces risk aversion) in the Balloon Analogue Risk Task and is associated with increased performance in the prehunch/hunch period of the Iowa Gambling Task. In Study 2, we found that naturally occurring negative emotions also increase risk aversion in Balloon Analogue Risk Task, but the incidental use of cognitive reappraisal of emotions impedes this effect. We offer evidence that the increased effectiveness of cognitive reappraisal in reducing the experience of emotions underlies its beneficial effects on decision making.

Keywords: emotion regulation, cognitive reappraisal, expressive suppression, decision making, risk aversion

It is well established that emotion plays a key role in human social and economic decision making (see, e.g., Elster, 1998; Loewenstein, 2000; Peters, Västfjäll, Gärling, & Slovic, 2006). People evaluate objective features of alternatives such as expected return in a subjective way (Edwards, 1962; Kahneman & Tversky, 1979), and emotions are understood to influence these subjective evaluations (Loewenstein & O'Donoghue, 2004; Naqvi, Shiv, & Bechara, 2006; Slovic, Finucane, Peters, & MacGregor, 2007). The recent literature on emotion regulation (ER), however, highlights that humans typically make efforts to control emotion experiences (Gross, 2002). This leaves open the possibility that decision effects attributed to acute emotions may be mediated by ER strategies. If so, this raises the additional possibility that different regulation strategies could have different decision implications. Only very recently, however, have scholars begun to investigate these possibilities. We report data from both designed and naturally occurring environments, providing convergent evidence that ER strategies modulate decision making. In particular,

we find that the decision effects of emotion vary according to the way in which a person regulates the emotion experience.

Various theoretical approaches have indicated that, contrary to traditional thinking in psychology and economics (Neisser, 1967; Simon, 1956), emotions play an active role in some forms of decision making. Regardless of whether they have been assimilated to the “goodness” or “badness” of alternatives for action (Slovic et al., 2007), attributed to activation in basic appetitive or defensive motivational systems (Bradley & Lang, 2007; Loewenstein & O'Donoghue, 2004), or reduced to somatic markers associated with current or past behavioral outcomes (Bechara, Damasio, & Damasio, 2000), emotions have been consistently shown to influence decision making.

In the emerging neuroeconomics literature, brain lesion, functional neuroimaging, and neurophysiological studies in animal models and humans have begun to shed light on the neural foundation of emotion and decision (Coricelli, Dolan, & Sirigu, 2007; O'Doherty & Bossaerts, 2008; Rangel, Camerer, & Montague, 2008; Seymour & Dolan, 2008). These studies suggest that humans can anticipate the emotional impact of potential future decisions using processes that involve the amygdala as well as the ventromedial prefrontal cortex (Bechara, Damasio, Damasio, & Lee, 1999; De Martino, Kumaran, Seymour, & Dolan, 2006; Weller, Levin, Shiv, & Bechara, 2007). This type of anticipation can be adaptive in that emotions such as anxiety or disgust have been shown to impair decision making (Lerner, Small, & Loewenstein, 2004; Preston, Buchanan, Stansfield, & Bechara, 2007), even when physiological responses properly signal disadvantageous alternatives (Miu, Heilman, & Houser, 2008).

The intrinsic role of emotion in decision is all the more important as the value of prospects (i.e., actions with uncertain rewards) is computed in “emotion–cognition brain hubs” (Pessoa, 2008)

Renata M. Heilman, Liviu G. Crişan, Mircea Miclea, and Andrei C. Miu, Emotion and Cognition Neuroscience Laboratory, Program of Cognitive Neuroscience, Department of Psychology, Babeş-Bolyai University; and Daniel Houser, Interdisciplinary Center for Economic Science and Department of Economics, George Mason University.

We thank Camelia M. Kuhn (Northwestern University) and Sergiu P. Paşca (Stanford University) and two anonymous reviewers for useful comments on a version of this article. We are grateful to the International Foundation for Research in Experimental Economics CNCSIS for supporting this research.

Correspondence concerning this article should be addressed to Andrei C. Miu, 37 Republicii, Cluj-Napoca, CJ 400015, România. E-mail: andrei_miu@emcoglab.org

such as midbrain dopaminergic regions and their targets (i.e., ventral and dorsal striatum, ventromedial and ventrolateral prefrontal cortex, anterior cingulate cortex). Therefore, neuroeconomists have emphasized that the interaction of emotion and decision making is profitably studied in environments that include risk (where the decision maker has perfect information regarding the stochastic relationship between actions and outcomes) and uncertainty (where the decision maker does not have full information about the stochastic environment; see, e.g., Rangel et al., 2008).

When a human anticipates or experiences an emotion, s/he will often use strategies to control that experience. It follows that ER, a concept subsuming the processes controlling which emotions we have, when we have them, and how we experience and express them (Gross, 2002), could be crucial to decision making as well as other cognitive processes (e.g., memory; Richards & Gross, 1999, 2000). Although many ER strategies may be in use (e.g., Garnefski, Legerstee, Kraaij, Van Den Kommer, & Teerds, 2002), most recent research has focused on more commonly used ER strategies that act either before emotions become activated (*antecedent-focused ER*) or after emotions arise (*response-focused ER*; Gross & Thompson, 2007). The process model that distinguishes ER strategies according to when they act in the emotion generative process (Gross, 1998b, 2002) has been exceptionally useful to cognitive scientists and generated pioneering discoveries on the effects of ER on cognition, physiology, and health (for recent reviews, see Canli, Ferri, & Duman, 2009; Gross, 2008; Phillips, Ladouceur, & Drevets, 2008).

Two specific ER strategies have been extensively investigated over the past decade (Ochsner & Gross, 2005, 2008). One of these, *cognitive reappraisal*, is an antecedent-focused ER strategy that alters the trajectory of emotional responses by reformulating the meaning of the situation. The other strategy, *expressive suppression*, is a response-focused strategy that involves inhibiting behaviors (e.g., facial expressions, verbal utterances, gestures) associated with emotional responding (Gross, 2002; Gross & Thompson, 2007). Whereas both reappraisal and suppression decrease the expression of emotions, their effectiveness in decreasing the experience of emotion differs due to their timing with respect to the emotion generative process. In particular, reappraisal and suppression effectively decrease the experience of positive emotions, but only the former is as effective in reducing the experience of negative emotions (Gross, 1998a; Gross & Levenson, 1997). In addition, reappraisal diminishes emotion at an early stage and without the need for sustained effort over time, whereas suppression involves active efforts to inhibit prepotent emotional responses (Gross, 2002; Muraven, Tice, & Baumeister, 1998). By way of increased effort or “ego depletion” (Baumeister, 2003), suppression impairs explicit memory (Richards & Gross, 1999, 2000). Similarly, suppression of negative emotions (e.g., anger, embarrassment, but not sadness) accompanied by high arousal is associated with impulsive decision making (Leith & Baumeister, 1996).

Recent research suggests that reappraisal and suppression might affect decision making differently. For instance, habitual use of cognitive reappraisal accounts for over half of the variance in the profit of simulated economic negotiations (Yurtsever, 2004, 2008). Similarly, individual differences in cognitive reappraisal and expressive suppression predict health risk behaviors (Magar, Phillips, & Hosie, 2008). The effects of reappraisal and suppression on

decision making, however, have not previously been directly compared.

We suggest that there may be two key routes by which reappraisal and suppression might generate different consequences for decision making. One, an “emotional” route, stems from differences between reappraisal and suppression in their effectiveness in mitigating the experience of negative and positive emotions (Gross, 2002). The second, a “nonemotional” route, stems from differences in the level of effort (cognitive load) required to implement reappraisal or suppression, which could perhaps be related to differences between their respective contributions to ego depletion (Baumeister, 2003; Richards & Gross, 1999).

Regarding the emotional route, recent studies offer evidence that the acute use of reappraisal effectively decreases physiological arousal related to the anticipation of reward (Delgado, Gillis, & Phelps, 2008) and loss aversion (Sokol-Hessner et al., 2009). The nonemotional route is emphasized in studies of self-regulation, including evidence that the expressive suppression of emotion modulates risk taking through ego depletion (Leith & Baumeister, 1996; Muraven et al., 1998).

In this research, we manipulated ER strategies under controlled conditions to provide rigorous evidence on their decision-making effects. We focused on instructed and incidental reappraisal and suppression used to regulate negative emotions induced by movies (Study 1) and naturally occurring negative and positive emotions (Study 2). We evaluated the effects of these ER strategies on decision making under both risk and uncertainty.

We studied decisions in the Balloon Analogue Risk Task (BART; Lejuez et al., 2002) as well as the Iowa Gambling Task (IGT; Bechara, Damasio, Damasio, & Anderson, 1994). BART is a computer-based measure of risk taking, in which participants can earn financial rewards by pumping balloons presented on a screen; different balloons have variable explosion points, and once a balloon explodes, the money deposited for pumping that balloon is lost (Lejuez et al., 2002). The other economic game, IGT, simulates real-life decision making in the way it factors uncertainty of premises and outcomes as well as reward and punishment (Bechara et al., 1994). It measures the degree to which individuals come to choose small immediate gains (associated in the long term with smaller losses) over large immediate gains (associated in the long term with larger losses; Bechara et al., 1994).

We next report (a) whether manipulations of ER of two negative emotions induced by films influence decision-making performance, and (b) whether the incidental use of ER strategies on naturally occurring positive and negative emotions affects decisions. We found statistically identical decision patterns between those who use suppression and a control group using no ER strategy. In contrast, in relation to both the control and suppression groups, participants using reappraisal displayed systematically and statistically different decision patterns. In particular, we found that reappraisal promotes increased risk taking (or equivalently, reduces risk aversion), and we traced the source of this effect to the emotional route.

Study 1

This study was designed to investigate the effects of ER on decision making in laboratory conditions that involved the controlled induction of two negative emotions. Fear and disgust were

induced by movies, and the effects of suppressing or reappraising these emotions were tested in IGT and BART. We chose to study fear and disgust for three reasons. First, they are both negative emotions that, according to the previous literature on ER (e.g., Gross & Levenson, 1997), should be effectively downregulated by reappraisal, but not suppression. Therefore, this study allowed us to test the emotional route by which ER may influence decision making. Second, little is known about the way different emotions of the same valence differentially influence judgment and choices (Lerner & Keltner, 2000). This is relevant because although both fear and disgust involve unpleasant states, they differ in important dimensions including certainty and attentional activity (Smith & Ellsworth, 1985). In contrast to disgust, which involves high levels of certainty and is associated with a strong unwillingness to attend to the situation, fear involves maximal uncertainty reflected in the apparent indecisiveness regarding whether to attend to the situation. In addition, disgust may have evolved to ward off contamination, and this might increase its effects on risk attitudes (Fessler, Pillsworth, & Flamson, 2004). Third, fear and disgust are among the emotions that can be reliably induced using movies in the laboratory (Kreibig, Wilhelm, Roth, & Gross, 2007; Rottenberg, Ray, & Gross, 2007).

In summary, laboratory-induced fear and disgust can plausibly inform the effects of ER on decision making in controlled conditions. In light of the known differences in their ability to downregulate negative emotions, we hypothesized that participants using reappraisal would make riskier decisions than those using suppression.

Method

Sixty participants (56 women; mean age = 21.45 years) from the Babeş-Bolyai University campus volunteered for this study. They were randomly distributed in six groups defined by the induced emotion (i.e., fear or disgust) and ER strategy (i.e., cognitive reappraisal, expressive suppression, or control/no ER instructions). Immediately prior to the experiment, participants completed the Positive and Negative Affect scales of the Positive and Negative Affect Schedule—Expanded Form (PANAS-X; Watson & Clark, 1999) to control for their dispositional mood. They also completed the Specific Affect scales of the same questionnaire as a pretest measure of emotion. Then, according to the experimental condition, the participants viewed one of two short (i.e., approximately 2 min) movie excerpts (either *Gothika*, 2003, or *Pink Flamingos*, 1972), both of which reliably elicit fear or disgust (Rottenberg et al., 2007).

Participants viewed the movies with standard instructions to reappraise, suppress their emotions, or in the absence of instructions related to ER (see also Richards & Gross, 2000). Immediately after the movie, they completed the Specific Affects scales of PANAS-X (posttest), enabling us to measure the induced emotion. They also completed the Emotion Regulation Questionnaire (ERQ; Gross & John, 2003) to confirm that participants employed the instructed ER strategy.

Next, participants played computer versions of BART and IGT. The average number of pumps per unexploded balloon in BART and the difference between advantageous (i.e., C and D) and disadvantageous (i.e., A and B) selections in IGT were the dependent variables, as described in the standard protocols (Bechara et

al., 1994; Lejuez et al., 2002). In addition, CD – AB scores were calculated for each 20-selection block of IGT, in light of the observation that these blocks reflect stages of decision-making optimization, which can be differentially influenced by emotions (see Bechara, Damasio, Tranel, & Damasio, 1997). The order of BART and IGT was counterbalanced between the two groups.

Results

Manipulation checks. The Positive and Negative Affect scores indicated that immediately before the movies, there were no significant differences in dispositional mood between the conditions. The movies succeeded in specifically inducing fear and disgust, as indicated by the significant increases in the corresponding specific affect scores in the posttest over all the groups (see Table 1). ERQ scores confirmed that the participants successfully used the strategy that they were instructed to use during the movies (see Table 1).

Emotion experience. The type of ER that participants used during the movie affected the emotion they experienced. In comparison to the control group, reappraisers, but not suppressors, displayed significantly decreased fear and disgust (see Table 1).

Decision-making performance. A 3 (ER: reappraisal vs. suppression vs. control) $\times$ 2 (emotion: fear vs. disgust) analysis of variance (ANOVA) indicated a significant main effect of ER on BART performance, $F(2, 57) = 22.69, p < .01, \eta^2 = .8$. The effects of Emotion or ER $\times$ Emotion on BART were not statistically significant. Post hoc analyses showed that reappraisers had significantly higher mean pumps per unexploded balloons in BART than suppressors and controls (see Figure 1A).

Similar statistical analyses on the total CD – AB scores indicated no significant effects of ER or emotion on IGT performance. However, by including the block of selections as an additional factor in the ANOVA, we found a significant effect of ER, $F(2, 57) = 19.8, p < .05, \eta^2 = .4$, and block, $F(4, 55) = 9.5, p < .01, \eta^2 = .3$, on IGT performance. Post hoc analyses indicated that reappraisers had significantly higher scores in the third block of trials compared with suppressors and controls (see Figure 2A).

Study 2

This experiment was designed to replicate the effects of ER on decision making in conditions that involved naturally occurring emotions and extend these effects to positive emotions. It is widely accepted that naturally occurring emotions are more salient and valenced than those induced in the laboratory, and their influences on cognition may be more conspicuous (for various perspectives, see Bradley & Lang, 2007; Damasio, 2005; Frijda, 1988). Fortunately, intense emotions such as those triggered by having just learned the result of a final exam can be measured and studied (see also Gramzow, Willard, & Mendes, 2008). We investigated the effects of incidental ER on decision making in this natural context.

Positive emotions offered a ground for testing whether the effects of ER on decision making are supported by the different degrees of effortfulness or cognitive load. Specifically, both reappraisal and suppression downregulate positive emotions (Gross & Levenson, 1997). In this context, any difference between the effects of these two ER strategies on decision making may be attributed to the increased effort that is required to suppress the

Table 1
Mood Before the Experiment, Specific Emotions Induced by Movies, and Use of Cognitive Reappraisal and Expressive Suppression in Study 1

Measure	Group (ER & emotion)	Mood		Specific affect				Cognitive reappraisal	Expressive suppression
		Positive affect	Negative affect	Pretest		Posttest			
				Fear	Disgust	Fear	Disgust		
Total	Fear	32.42 ± 5.6	22.18 ± 7.86	10.19 ± 3.94	8.17 ± 3.5	12.27 ± 4.69**	8.16 ± 3.72	3.77 ± 1.46	2.53 ± 1.1
	Disgust	35.36 ± 5.58	21.8 ± 6.71	10.19 ± 3.94	8.17 ± 3.5	10.47 ± 4.83	9.31 ± 4.03**	4.00 ± 1.52	2.47 ± 1.06
	Fear	34.44 ± 4.64	21 ± 8.18	10 ± 4.53	8.55 ± 4.3	10.15 ± 4.41	8.44 ± 4.36	5.68 ± 0.16**	1.77 ± 0.19
	Disgust	38.45 ± 4	20.18 ± 4.97	9.54 ± 3.11	8.5 ± 4.13	9.5 ± 2.94	8.4 ± 4.01	5.95 ± 0.03**	1.81 ± 0.31
Suppressors	Fear	30.71 ± 7.29	20.87 ± 8.7	10.27 ± 3.49	9.37 ± 5.34	12.8 ± 4.95**	9.31 ± 4.59	2.81 ± 0.13	4 ± 0.65**
	Disgust	33.8 ± 6.9	22.2 ± 6.49	8.7 ± 2.16	8.27 ± 3.92	8.7 ± 2.05	11.02 ± 4.27**	2.91 ± 0.16	3.82 ± 0.67**
	Fear	31.8 ± 5.71	24.3 ± 7.24	10.31 ± 3.9	7.6 ± 2.17	14 ± 4.05**	7.6 ± 2.31	2.83 ± 0.07	2.05 ± 0.62
	Disgust	33.33 ± 4.3	23.33 ± 8.86	9.88 ± 4.22	7.73 ± 2.28	9.44 ± 3.57	10.42 ± 3.15**	2.83 ± 0.19	1.77 ± 0.29

Note. ER = emotion regulation.

** $p < .01$ (Student t tests) in comparison to the corresponding pretest measure. ** $p < .01$ (ANOVA) in comparison to the other two ER groups. Values are means ± 1 standard deviation.

expression of emotions. Therefore, this experiment enabled us to replicate and extend the previous results to naturally occurring negative and positive emotions, as well as to determine whether the effects of ER on decision making are supported by emotional or nonemotional mechanisms.

Again, our hypothesis was that the reappraisal of negative emotions would reduce risk aversion in comparison to the suppression of negative emotions. We also expected to observe no differences in risk attitudes between participants using reappraisal or suppression to control positive emotions.

Method

Forty-four students (33 women; mean age = 22.6 years) attending an undergraduate course at Babeş-Bolyai University agreed to participate in this study. All the participants had the final exam several days before the experiment, and on that occasion they had been asked to provide self-evaluations of their performance and to assess the importance they placed on that exam. The participants for this study were recruited from those students for whom the exam was extremely important and either under- or overestimated their evaluation by 1 or 2 points (of 10). Immediately after learning their exam results, they were asked to volunteer for this study, and those who agreed completed the PANAS-X. They also completed ERQ to identify those who incidentally used reappraisal, suppression, or neither of these ER strategies to control the emotions elicited by learning their exam outcome. Just as in the above-discussed experiment, immediately after completing the questionnaires, participants were taken to a nearby laboratory room where they played BART and IGT.

Results

Manipulation checks. An analysis of covariance confirmed our prediction that self-evaluations of the exam results (under- vs. overestimation) would have a significant effect on negative, $F(1, 42) = 4.3, p < .05, \eta^2 = .4$, and positive affect scores, $F(1, 42) = 7.83, p < .01, \eta^2 = .6$, with the differences between the self- and the objective evaluations included as a covariate. Post hoc analyses indicated that the participants who overestimated their evaluation displayed more negative affect than those who underestimated their evaluation, and the latter displayed more positive affect than the former (see Table 2). ERQ scores indicated that 17 participants (9 who overestimated their evaluation) predominantly used reappraisal, 13 participants (7 who overestimated) used suppression, and 14 (7 who overestimated) used neither reappraisal nor suppression (hereinafter “controls”) to regulate emotions elicited by learning their exam score (see Table 2).

Emotion experience. The type of ER used by participants had a significant effect on positive, $F(2, 41) = 5.6, p < .01, \eta^2 = .7$, and negative affect, $F(2, 41) = 4.63, p < .05, \eta^2 = .5$. In comparison to controls, reappraisors, but not suppressors, reported reduced negative affect. However, both reappraisors and suppressors had significantly reduced positive affect scores compared with controls (see Table 2).

Decision-making performance. A 3 (ER: reappraisal vs. suppression vs. control) $\times$ 2 (emotion: positive vs. negative affect) ANOVA indicated the significant main effect of emotion, $F(1, 42) = 16.8, p < .01, \eta^2 = .9$, and the significant interaction of

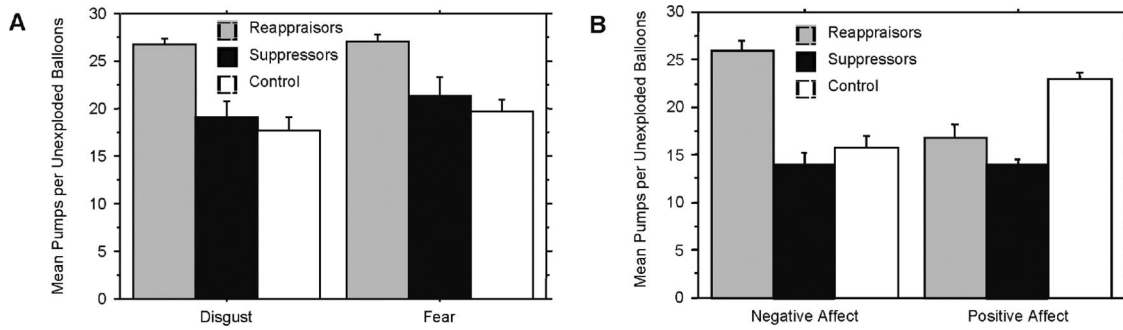


Figure 1. Balloon Analogue Risk Task performance in Study 1 (A) and Study 2 (B). Error bars indicate standard error of the mean.

ER $\times$ Emotion on BART performance, $F(4, 39) = 8.13, p < .01, \eta^2 = .7$. Post hoc analyses showed that negative affect reduced risk taking in comparison to positive affect. In addition, in relation to suppressors, participants who reappraised their negative affect displayed increased risk taking (see Figure 1B).

Similar statistical analyses identified a significant interaction of ER $\times$ Emotion $\times$ Block on IGT performance, $F(9, 34) = 3.7, p < .01, \eta^2 = .5$. Post hoc tests indicated that participants who reappraised their negative affect displayed increased CD – AB scores in the third block of IGT, in comparison to the suppressors and controls, who also experienced negative affect (see Figure 2B).

General Discussion

The present results indicate that acute cognitive reappraisal increases risk taking by effectively reducing the experience of negative emotions. In contrast, expressive suppression does not decrease risk aversion because it is ineffective in regulating unpleasant feelings. These effects were replicated in conditions that involved fear and disgust induced by movies in the laboratory, as well as naturally occurring negative emotions triggered by learning the previously overestimated result of an exam. The effects of ER on decision making were also investigated in conditions that involved naturally occurring positive emotions, where there were no differences in risk taking as a function of ER. This may indicate that in conditions (i.e., positive emotions) in which reappraisal and

suppression are equally effective in reducing the experience of emotion, the difference in effortfulness between them does not further influence decision making. Therefore, these results suggest that ER affects decision making under risk and uncertainty by reducing the experience of emotion.

Previous studies showed that negative emotions such as anxiety increase risk aversion in BART (Maner et al., 2007) and impair decision-making optimization in IGT (Miu, Heilman, et al., 2008; Preston et al., 2007). Experimentally induced and naturally occurring fear is also associated with risk aversion and pessimistic risk appraisals (Lerner & Keltner, 2001; see also Gasper & Clore, 1998; Miu, Micalea, & Houser, 2008; Stöber, 1997). The magnitude of risk aversion induced by fear and disgust in the present study (i.e., the control conditions in Study 1) was similar to that reported in previous studies using BART (e.g., Maner et al., 2007). However, the aim of the present Study 1 was to investigate the effects of regulating the experience of these emotions using reappraisal or suppression on decision making under risk. We show for the first time that instructing participants to reappraise both fear and disgust significantly reduces the unpleasant experience of these emotions and consequently promotes risky decision making (or decreases risk aversion) in relation to both a control group as well as a group using suppression.

This effect was replicated in Study 2 in which participants incidentally used reappraisal to reduce the unpleasant feelings

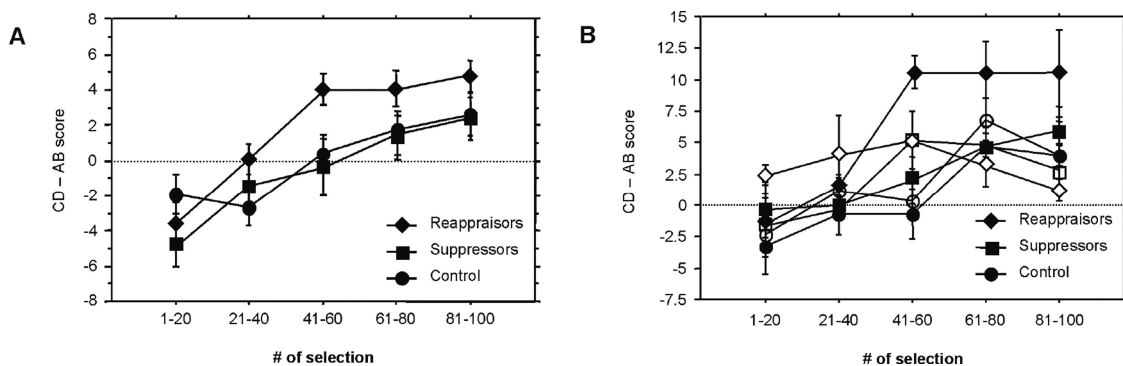


Figure 2. Iowa Gambling Task performance in Study 1 (A) and Study 2 (B). In (B), solid symbols are for the negative affect condition, and open symbols are for the positive affect condition. Error bars indicate standard error of the mean.

Table 2

Positive and Negative Affect Induced by Learning the Results of the Exam, and Use of Cognitive Reappraisal and Expressive Suppression in Study 2

Measure	Group (ER & result anticipation)	General positive affect	General negative affect	Cognitive reappraisal	Expressive suppression
Total	Underestimation	35.58 ± 15.15	27.58 ± 3.2	4.77 ± 1.22	2.87 ± 1.22
	Overestimation	27.75 ± 3.27	43.12 ± 14.52**	4.61 ± 1.3	3 ± 1.53
Reappraisors	Underestimation	27.5 ± 4.35	27 ± 3.16	6.29 ± 0.64 ^{§§}	2 ± 0.35
	Overestimation	30.2 ± 1.48	26.4 ± 2.3	6.36 ± 0.59 ^{§§}	2.1 ± 0.28
Suppressors	Underestimation	27.4 ± 3.78	25.8 ± 2.28	3.76 ± 0.4	4.2 ± 0.32 ^{§§}
	Overestimation	25.33 ± 3.07	53.33 ± 11.79**	3.77 ± 0.2	4.7 ± 1.05 ^{§§}
Controls	Underestimation	60 ± 3.6	31.33 ± 1.15	4.44 ± 0.25	1.83 ± 0.52
	Overestimation	28.2 ± 3.11**	47.6 ± 9.01*	3.86 ± 0.61	1.85 ± 0.57

Note. ER = emotion regulation.

* $p < .05$ and ** $p < .01$ (ANOVA) in comparison to the group that underestimated the exam result. ^{§§} $p < .01$ (ANOVA) in comparison to the other two ER groups. Values are means ± 1 standard deviation.

triggered by learning the previously overestimated result of an important exam and consequently displayed reduced risk aversion in BART. By effectively downregulating the experience of negative emotions, reappraisal contributes to an increased sense of emotional control that mitigates aversion to risky decisions. This interpretation accords well with previous results suggesting that the appraisal of control mediates the relationship between emotions and risk taking (Lerner & Keltner, 2001).

In contrast to reappraisal, suppression is ineffective in reducing the experience of negative emotions. Consequently, risk attitudes among suppressors are not changed in relation to those experiencing negative emotions but not using ER strategies. Focusing on the ego-depleting capacity of self-regulation, several studies have argued that bad moods increase risk aversion, and the suppression of these moods may augment this effect (Leith & Baumeister, 1996; Muraven et al., 1998). However, these studies have focused on emotions (e.g., embarrassment) that differ in important dimensions of appraisal (e.g., control) and sometimes failed to induce bad moods that significantly differ from neutral moods (see, e.g., Study 2 in Leith & Baumeister, 1996). Our results offer compelling evidence that expressive suppression cannot mitigate risk aversion induced by negative emotions because it cannot decrease the experience of these emotions. At the same time, future studies might control for both the degree of effortfulness as well as the success of ER in reducing experienced emotions in order to compare the effects of ER on risk taking through emotional and nonemotional mechanisms.

A different mechanism may be involved in the effect of reappraisal on IGT. Previous studies have shown that the optimization of performance in IGT is influenced by somatic markers such as sympathetic increases in skin conductance that anticipate disadvantageous selections and adaptively bias decision making (Bechara et al., 1999). This role of somatic markers is especially important in the so-called “prehunch” period, when participants start to show a preference for advantageous decks without being yet able to tell why (Bechara et al., 1997). Negative emotions such as anxiety alter IGT performance by increasing the physiological noise that impedes somatic markers, along with the tendency to overwrite the adaptive influence of somatic markers through worry and rumination (Miu, Heilman, et al., 2008; Preston et al., 2007). Reappraisal decreases sympathetic responses related to anticipa-

tion of reward (Delgado et al., 2008) and loss aversion (Sokol-Hessner et al., 2009), whereas suppression increases sympathetic activation (Gross, 1998a; Gross & Levenson, 1997). Therefore, the opposing physiological effects of these two ER strategies probably support the increased performance of reappraisors compared with suppressors in the transition from the “prehunch” to the “hunch” period in this study. In future psychophysiological studies, we intend to directly investigate the effects of ER on somatic markers and declarative learning in IGT.

Turning to positive emotions, previous studies have shown that happiness reduces risk aversion (Lerner & Keltner, 2001). This view is also supported by the results of the present Study 2 that showed reduced risk aversion in a condition that involved pleasant feelings triggered by learning the previously underestimated results of an exam. We show that the incidental use of reappraisal and suppression to control these positive emotions abolishes their effect on risk aversion. This effect is explained by the similar effectiveness of both ER strategies on reducing the experience of positive emotions (Gross, 1998a, 2002; Gross & Levenson, 1997). The only difference that exists between reappraisal and suppression in this condition is the increased effort associated with the latter (Muraven et al., 1998). However, we found no difference in risk taking between the conditions that involved the reappraisal and suppression of positive emotions. The implication is that the differential effectiveness of these ER strategies in reducing the experience of emotion (what we denoted above as the “emotional route”) seems to be the primary mechanism underlying their effects on risk attitudes.

The present results may also be viewed through the lens of cognitive neuroscience and neuroeconomics. Several studies offer convergent evidence on the key role played by ventromedial prefrontal cortex (VMPFC) in reappraisal (Banks, Eddy, Angstadt, Nathan, & Phan, 2007; Goldin, McRae, Ramel, & Gross, 2008; Wager, Davidson, Hughes, Lindquist, & Ochsner, 2008). The functional connectivity between VMPFC and amygdala (Banks et al., 2007; Goldin et al., 2008), insula (Goldin et al., 2008), as well as nucleus accumbens (Wager et al., 2008) is perhaps connected to the success of reappraisal in reducing experienced emotions. It is intriguing that functional neuroimaging studies have shown that the recruitment of VMPFC is also central in IGT performance (Northoff et al., 2006). Indeed, its failure may result in altered risk

taking that characterizes the performance of patients with substance dependence or gambling problems (Tanabe et al., 2007).

Further evidence suggests that reappraisal may increase the efficacy of emotion–cognition “brain hubs” (Pessoa, 2008), such as prefrontal–amygdala and prefrontal–mesolimbic circuits that support optimal decision making under risk and uncertainty. In particular, the insula and basal ganglia may be specifically involved in the anticipation of outcomes in IGT (Lin, Chiu, Cheng, & Hsieh, 2008), and voluntary risk taking in BART is believed to involve the activation of a frontal–mesolimbic pathway (Rao, Korczykowski, Pluta, Hoang, & Detre, 2008).

The finding that reappraisal affects risk attitudes is also relevant to health psychology and psychiatry. On the one hand, a reduced risk aversion due to habitual use of reappraisal may contribute to an enhanced well-being among the general population (Gross & John, 2003; see also Magar et al., 2008). On the other hand, biased risk aversion has been involved in the pathogenesis of affective disorders (Chandler, Wakeley, Goodwin, & Rogers, 2009), as well as addiction (e.g., George, Rogers, & Duka, 2005) and pathological gambling (e.g., Brand et al., 2005). Poor ER skills and reduced ER-related neural activation have also been described in anxiety (Goldin, Manber, Hakimi, Canli, & Gross, 2009). Therefore, training these patients in effective ER strategies such as reappraisal may provide a means of both enhancing their emotional control as well as reducing decision-making biases.

The use of BART and IGT in this study had the advantages of task simplicity and increased interdisciplinary relevance due to their extensive applications in psychology, economics, and neuroscience. However, future studies of ER and decision making might use other risk-taking tasks (see, e.g., Holt & Laury, 2002; Tom, Fox, Trepel, & Poldrack, 2007), which would allow the estimation of utility functions that drive choices. This might be important to determine whether reappraisal has distinct effects on the valuation of losses and gains, and would be an important step toward connecting ER to prospect theory (see, e.g., Kahneman, 2003; Rangel et al., 2008). Ongoing investigations in our laboratory target the effects of ER on susceptibility to framing and cooperation in social decision making, highlighting the relevance of this line of research to economics. It would be profitable for future studies to replicate the present results using continuous measurements of the effects of ER on emotions experienced during the decision-making tasks. This could be accomplished using systems such as the Affect Rating Dial (see Ruef & Levenson, 2007).

In conclusion, the studies reported here show that cognitive reappraisal changes risk attitudes. By efficaciously downregulating a negative emotion experience, reappraisal enables riskier decision making that does not occur either with expressive suppression or an absent ER strategy. Emotion has taken the center stage in decision theory, and with it ER promises to play an increasingly prominent role in psychology, economics, and cognitive neuroscience.

References

- Banks, S. J., Eddy, K. T., Angstadt, M., Nathan, P. J., & Phan, K. L. (2007). Amygdala–frontal connectivity during emotion regulation. *Social Cognitive and Affective Neuroscience*, 2, 303–312.
- Baumeister, R. F. (2003). Ego depletion and self-regulation failure: A resource model of self-control. *Alcoholism: Clinical and Experimental Research*, 27(2), 1–4.
- Bechara, A., Damasio, A. R., Damasio, H., & Anderson, S. W. (1994). Insensitivity to future consequences following damage to human prefrontal cortex. *Cognition*, 50, 7–15.
- Bechara, A., Damasio, H., & Damasio, A. R. (2000). Emotion, decision making and the orbitofrontal cortex. *Cerebral Cortex*, 10, 295–307.
- Bechara, A., Damasio, H., Damasio, A. R., & Lee, G. P. (1999). Different contributions of the human amygdala and ventromedial prefrontal cortex to decision-making. *Journal of Neuroscience*, 19, 5473–5481.
- Bechara, A., Damasio, H., Tranel, D., & Damasio, A. R. (1997, February 28). Deciding advantageously before knowing the advantageous strategy. *Science*, 275, 1293–1295.
- Bradley, M. M., & Lang, P. J. (2007). Emotion and motivation. In J. T. Cacioppo, L. G. Tassinary, & G. G. Berntson (Eds.), *Handbook of psychophysiology* (3rd ed., pp. 581–607). New York: Cambridge University Press.
- Brand, M., Kalbe, E., Labudda, K., Fujiwara, E., Kessler, J., & Markowitsch, H. J. (2005). Decision-making impairments in patients with pathological gambling. *Psychiatry Research*, 133, 91–99.
- Canli, T., Ferri, J., & Duman, E. A. (2009). Genetics of emotion regulation. *Neuroscience*, 164, 43–54.
- Chandler, R. A., Wakeley, J., Goodwin, G. M., & Rogers, R. D. (2009). Altered risk-aversion and risk-seeking behavior in bipolar disorder. *Biological Psychiatry*, 66, 840–846.
- Coricelli, G., Dolan, R. J., & Sirigu, A. (2007). Brain, emotion and decision making: The paradigmatic example of regret. *Trends in Cognitive Sciences*, 11, 258–265.
- Damasio, A. R. (2005). *Descartes' error: Emotion, reason, and the human brain*. New York: Penguin.
- Delgado, M. R., Gillis, M. M., & Phelps, E. A. (2008). Regulating the expectation of reward via cognitive strategies. *Nature Neuroscience*, 11, 880–881.
- De Martino, B., Kumaran, D., Seymour, B., & Dolan, R. J. (2006, August 4). Frames, biases, and rational decision-making in the human brain. *Science*, 313, 684–687.
- Edwards, W. (1962). Subjective probabilities inferred from decisions. *Psychological Review*, 69, 109–135.
- Elster, J. (1998). Emotions and economic theory. *Journal of Economic Literature*, 36, 47–74.
- Fessler, D. M. T., Pillsworth, E. G., & Flanson, T. J. (2004). Angry men and disgusted women: An evolutionary approach to the influence of emotions on risk taking. *Organizational Behavior and Human Decision Processes*, 95, 107–123.
- Frijda, N. H. (1988). The laws of emotion. *American Psychologist*, 43, 349–358.
- Garnefski, N., Legerstee, J., Kraaij, V. V., Van Den Kommer, T., & Teerds, J. (2002). Cognitive coping strategies and symptoms of depression and anxiety: A comparison between adolescents and adults. *Journal of Adolescence*, 25, 603–611.
- Gasper, K., & Clore, G. L. (1998). The persistent use of negative affect by anxious individuals to estimate risk. *Journal of Personality and Social Psychology*, 74, 1350–1363.
- George, S., Rogers, R. D., & Duka, T. (2005). The acute effect of alcohol on decision making in social drinkers. *Psychopharmacology (Berl)*, 182, 160–169.
- Goldin, P. R., Manber, T., Hakimi, S., Canli, T., & Gross, J. J. (2009). Neural bases of social anxiety disorder: Emotional reactivity and cognitive regulation during social and physical threat. *Archives of General Psychiatry*, 66, 170–180.
- Goldin, P. R., McRae, K., Ramel, W., & Gross, J. J. (2008). The neural bases of emotion regulation: Reappraisal and suppression of negative emotion. *Biological Psychiatry*, 63, 577–586.
- Gramzow, R. H., Willard, G., & Mendes, W. B. (2008). Big tales and cool heads: Academic exaggeration is related to cardiac vagal reactivity. *Emotion*, 8, 138–144.

- Gross, J. J. (1998a). Antecedent- and response-focused emotion regulation: Divergent consequences for experience, expression, and physiology. *Journal of Personality and Social Psychology*, 74, 224–237.
- Gross, J. J. (1998b). The emerging field of emotion regulation: An integrative review. *Review of General Psychology*, 2, 271–299.
- Gross, J. J. (2002). Emotion regulation: Affective, cognitive, and social consequences. *Psychophysiology*, 39, 281–291.
- Gross, J. J. (2008). Emotion regulation. In M. Lewis, J. M. Haviland-Jones, & L. F. Barrett (Eds.), *Handbook of emotions* (3rd ed., pp. 497–512). New York: Guilford Press.
- Gross, J. J., & John, O. P. (2003). Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *Journal of Personality and Social Psychology*, 85, 348–362.
- Gross, J. J., & Levenson, R. W. (1997). Hiding feelings: The acute effects of inhibiting negative and positive emotion. *Journal of Abnormal Psychology*, 106, 95–103.
- Gross, J. J., & Thompson, R. A. (2007). Emotion regulation: Conceptual foundations. In J. J. Gross (Ed.), *Handbook of emotion regulation* (pp. 3–24). New York: Guilford Press.
- Holt, C. A., & Laury, S. K. (2002). Risk aversion and incentive effects. *The American Economic Review*, 92, 1644–1655.
- Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58, 697–720.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47, 263–292.
- Kreibig, S. D., Wilhelm, F. H., Roth, W. T., & Gross, J. J. (2007). Cardiovascular, electrodermal, and respiratory response patterns to fear- and sadness-inducing films. *Psychophysiology*, 44, 787–806.
- Leith, K. P., & Baumeister, R. F. (1996). Why do bad moods increase self-defeating behavior? Emotion, risk taking, and self-regulation. *Journal of Personality and Social Psychology*, 71, 1250–1267.
- Lejuez, C. W., Read, J. P., Kahler, C. W., Richards, J. B., Ramsey, S. E., Stuart, G. L., . . . Brown, R. A. (2002). Evaluation of a behavioral measure of risk taking: The Balloon Analogue Risk Task (BART). *Journal of Experimental Psychology: Applied*, 8, 75–84.
- Lerner, J. S., & Keltner, D. (2000). Beyond valence: Toward a model of emotion-specific influences on judgment and choice. *Cognition & Emotion*, 14, 473–493.
- Lerner, J. S., & Keltner, D. (2001). Fear, anger, and risk. *Journal of Personality and Social Psychology*, 81, 146–159.
- Lerner, J. S., Small, D. A., & Loewenstein, G. (2004). Heart strings and purse strings: Carryover effects of emotions on economic decisions. *Psychological Science*, 15, 337–341.
- Lin, C. H., Chiu, Y. C., Cheng, C. M., & Hsieh, J. C. (2008). Brain maps of Iowa Gambling Task. *BMC Neuroscience*, 9, 72.
- Loewenstein, G. (2000). Emotions in economic theory and economic behavior. *American Economic Review: Papers and Proceedings*, 90, 426–432.
- Loewenstein, G., & O'Donoghue, T. (2004). *Animal spirits: Affective and deliberative processes in economic behavior*. Retrieved from <http://cbdr.cmu.edu/seminar/Loewenstein.pdf>
- Magar, E. C. E., Phillips, L. H., & Hosie, J. A. (2008). Self-regulation and risk-taking. *Personality and Individual Differences*, 45, 153–159.
- Maner, J. K., Richey, J. A., Cromer, K., Mallott, M., Lejuez, C. W., Joiner, T. E., & Schmidt, N. B. (2007). Dispositional anxiety and risk-avoidant decision-making. *Personality and Individual Differences*, 42, 665–675.
- Miu, A. C., Heilman, R. M., & Houser, D. (2008). Anxiety impairs decision-making: Psychophysiological evidence from an Iowa Gambling Task. *Biological Psychology*, 77, 353–358.
- Miu, A. C., Miclea, M., & Houser, D. (2008). Anxiety and decision-making: Toward a neuroeconomics perspective. *Advances in Health Economics and Health Services Research*, 20, 55–84.
- Muraven, M., Tice, D. M., & Baumeister, R. F. (1998). Self-control as limited resource: Regulatory depletion patterns. *Journal of Personality and Social Psychology*, 74, 774–789.
- Naqvi, N., Shiv, B., & Bechara, A. (2006). The role of emotion in decision making: A cognitive neuroscience perspective. *Current Directions in Psychological Science*, 15, 260–264.
- Neisser, U. (1967). *Cognitive psychology*. Englewood Cliffs, NJ: Prentice Hall.
- Northoff, G., Grimm, S., Boeker, H., Schmidt, C., Bermpohl, F., Heinzl, A., . . . Boesiger, P. (2006). Affective judgment and beneficial decision making: Ventromedial prefrontal activity correlates with performance in the Iowa Gambling Task. *Human Brain Mapping*, 27, 572–587.
- Ochsner, K. N., & Gross, J. J. (2005). The cognitive control of emotion. *Trends in Cognitive Sciences*, 9, 242–249.
- Ochsner, K. N., & Gross, J. J. (2008). Cognitive emotion regulation: Insights from social cognitive and affective neuroscience. *Current Directions in Psychological Science*, 17, 153–158.
- O'Doherty, J. P., & Bossaerts, P. (2008). Toward a mechanistic understanding of human decision making. Contributions of functional neuroimaging. *Current Directions in Psychological Science*, 17, 119–123.
- Pessoa, L. (2008). On the relationship between emotion and cognition. *Nature Reviews Neuroscience*, 9, 148–158.
- Peters, E., Västfjäll, D., Gärling, T., & Slovic, P. (2006). Affect and decision making: A “hot” topic. *Journal of Behavioral Decision Making*, 19, 79–85.
- Phillips, M. L., Ladouceur, C. D., & Drevets, W. C. (2008). A neural model of voluntary and automatic emotion regulation: Implications for understanding the pathophysiology and neurodevelopment of bipolar disorder. *Molecular Psychiatry*, 13, 833–857.
- Preston, S. D., Buchanan, T. W., Stansfield, R. B., & Bechara, A. (2007). Effects of anticipatory stress on decision making in a gambling task. *Behavioral Neuroscience*, 121, 257–263.
- Rangel, A., Camerer, C., & Montague, P. R. (2008). A framework for studying the neurobiology of value-based decision making. *Nature Reviews Neuroscience*, 9, 545–556.
- Rao, H., Kordzykowski, M., Pluta, J., Hoang, A., & Detre, J. A. (2008). Neural correlates of voluntary and involuntary risk taking in the human brain: An fMRI study of the Balloon Analog Risk Task (BART). *NeuroImage*, 42, 902–910.
- Richards, J. M., & Gross, J. J. (1999). Composure at any cost? The cognitive consequences of emotion suppression. *Personality and Social Psychology Bulletin*, 25, 1033–1044.
- Richards, J. M., & Gross, J. J. (2000). Emotion regulation and memory: The cognitive costs of keeping one's cool. *Journal of Personality and Social Psychology*, 79, 410–424.
- Rottenberg, J., Ray, R. R., & Gross, J. J. (2007). Emotion elicitation using films. In J. A. Coan & J. J. B. Allen (Eds.), *Handbook of emotion elicitation and assessment* (pp. 9–28). New York: Oxford University Press.
- Ruef, A. M., & Levenson, R. W. (2007). Continuous measurement of emotion: The Affect Rating Dial. In J. A. Coan & J. J. B. Allen (Eds.), *Handbook of emotion elicitation and assessment* (pp. 286–297). New York: Oxford University Press.
- Seymour, B., & Dolan, R. (2008). Emotion, decision making, and the amygdala. *Neuron*, 58, 662–671.
- Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review*, 63, 129–138.
- Slovic, P., Finucane, M. L., Peters, E., & MacGregor, D. G. (2007). The affect heuristic. *European Journal of Operational Research*, 177, 1333–1352.
- Smith, C. A., & Ellsworth, P. C. (1985). Patterns of cognitive appraisal in emotion. *Journal of Personality and Social Psychology*, 48, 813–838.
- Sokol-Hessner, P., Hsu, M., Curley, N. G., Delgado, M. R., Camerer, C. F., & Phelps, E. A. (2009). Thinking like a trader selectively reduces

- individuals' loss aversion. *Proceedings of the National Academy of Sciences, USA*, 106, 5035–5040.
- Stöber, J. (1997). Trait anxiety and pessimistic appraisal of risk and chance. *Personality and Individual Differences*, 22, 465–476.
- Tanabe, J., Thompson, L., Claus, E., Dalwani, M., Hutchison, K., & Banich, M. T. (2007). Prefrontal cortex activity is reduced in gambling and nongambling substance users during decision-making. *Human Brain Mapping*, 28, 1276–1286.
- Tom, S. M., Fox, C. R., Trepel, C., & Poldrack, R. A. (2007, January 26). The neural basis of loss aversion in decision-making under risk. *Science*, 315, 515–518.
- Wager, T. D., Davidson, M. L., Hughes, B. L., Lindquist, M. A., & Ochsner, K. N. (2008). Prefrontal–subcortical pathways mediating successful emotion regulation. *Neuron*, 59, 1037–1050.
- Watson, D., & Clark, L. A. (1999). *PANAS-X. Manual for the Positive and Negative Affect Schedule—Expanded form*. Ames: The University of Iowa.
- Weller, J. A., Levin, I. P., Shiv, B., & Bechara, A. (2007). Neural correlates of adaptive decision making for risky gains and losses. *Psychological Science*, 18, 958–964.
- Yurtsever, G. (2004). Emotional regulation strategies and negotiation. *Psychological Reports*, 95(Pt. 1), 780–786.
- Yurtsever, G. (2008). Negotiators' profit predicted by cognitive reappraisal, suppression of emotions, misrepresentation of information, and tolerance of ambiguity. *Perceptual and Motor Skills*, 106, 590–608.

Received November 25, 2008

Revision received September 14, 2009

Accepted October 1, 2009 ■

Members of Underrepresented Groups: Reviewers for Journal Manuscripts Wanted

If you are interested in reviewing manuscripts for APA journals, the APA Publications and Communications Board would like to invite your participation. Manuscript reviewers are vital to the publications process. As a reviewer, you would gain valuable experience in publishing. The P&C Board is particularly interested in encouraging members of underrepresented groups to participate more in this process.

If you are interested in reviewing manuscripts, please write APA Journals at Reviewers@apa.org. Please note the following important points:

- To be selected as a reviewer, you must have published articles in peer-reviewed journals. The experience of publishing provides a reviewer with the basis for preparing a thorough, objective review.
- To be selected, it is critical to be a regular reader of the five to six empirical journals that are most central to the area or journal for which you would like to review. Current knowledge of recently published research provides a reviewer with the knowledge base to evaluate a new submission within the context of existing research.
- To select the appropriate reviewers for each manuscript, the editor needs detailed information. Please include with your letter your vita. In the letter, please identify which APA journal(s) you are interested in, and describe your area of expertise. Be as specific as possible. For example, “social psychology” is not sufficient—you would need to specify “social cognition” or “attitude change” as well.
- Reviewing a manuscript takes time (1–4 hours per manuscript reviewed). If you are selected to review a manuscript, be prepared to invest the necessary time to evaluate the manuscript thoroughly.